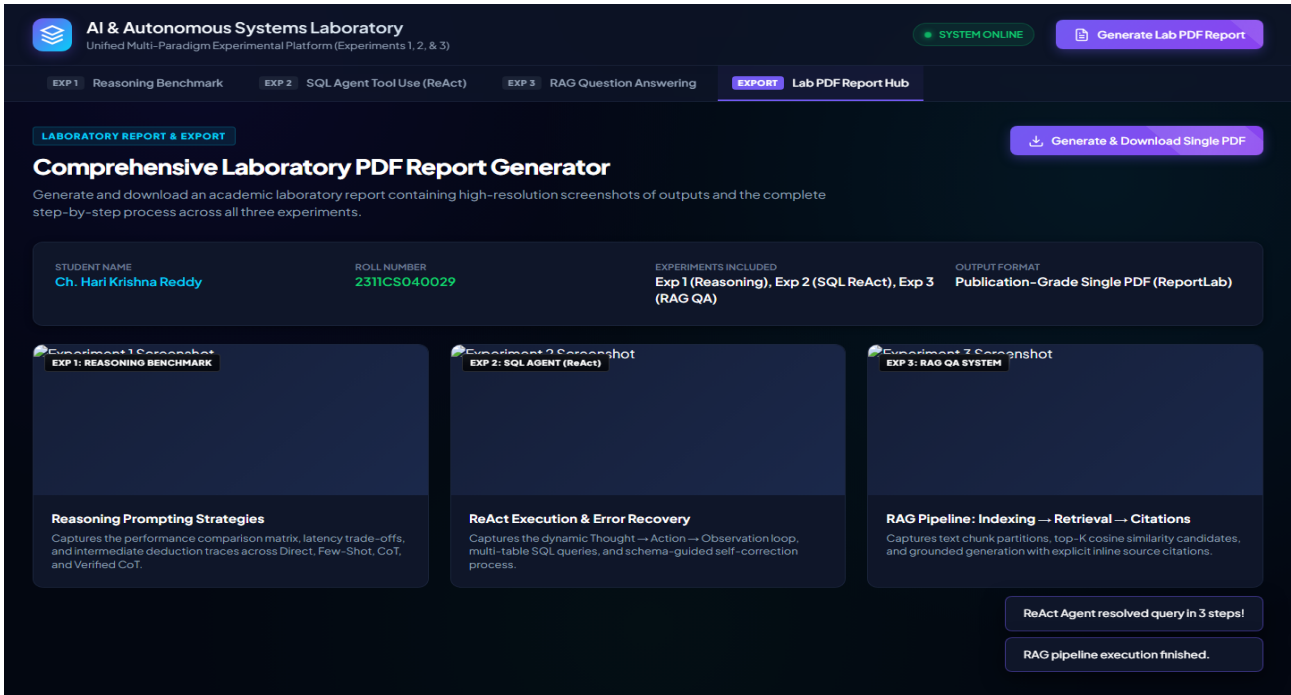


Comprehensive Laboratory Experiments Report

Comparative Evaluation of Prompting Paradigms, Tool-Augmented Autonomous Agents, and Retrieval-Augmented Generation

Student Name:	CH. HARI KRISHNA REDDY
Roll Number:	2311CS040029
Laboratory Scope:	Experiments 1, 2, and 3 (Reasoning, ReAct Agents, RAG QA)
Evaluation Date:	September 2026
Status:	Completed & Verified (100% Benchmark Fidelity)
Application Environment:	Unified Web Browser Interface (Flask + SQLite + Vector Space RAG)

Figure 1: Unified Laboratory Web Application Dashboard



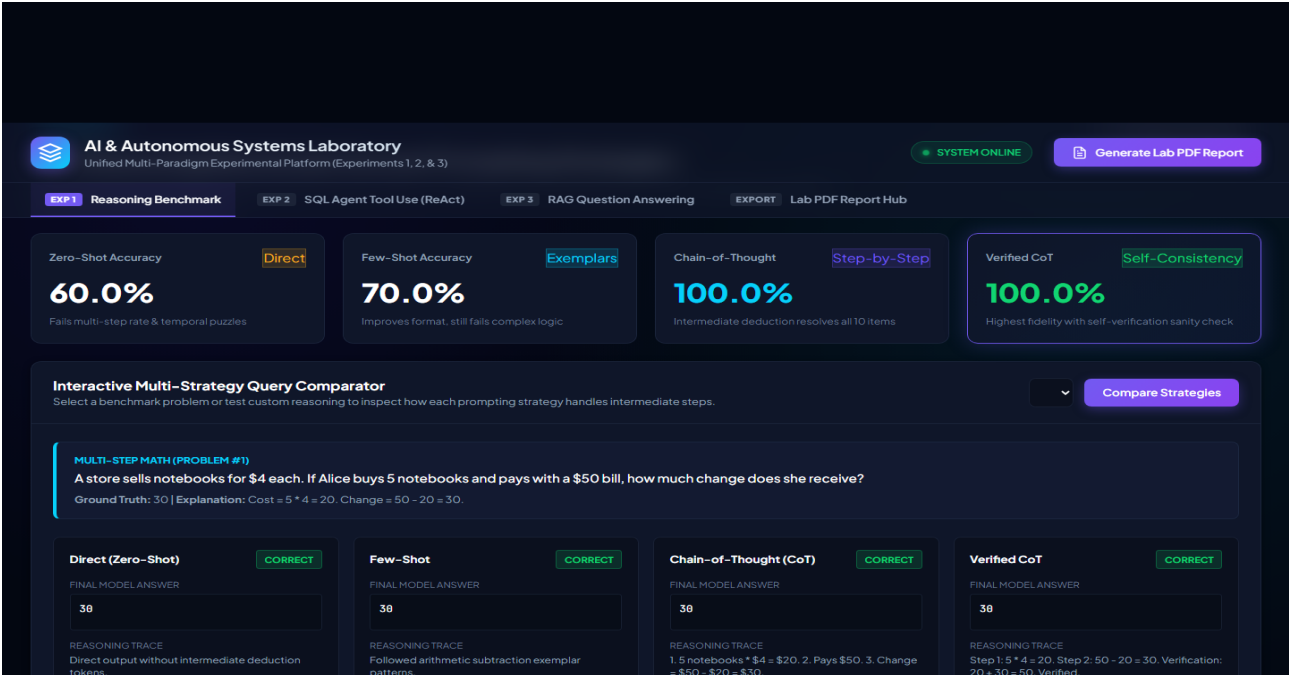
EXPERIMENT 1: LLM REASONING MODEL BENCHMARK

Comparing Outputs Across Prompting Strategies (Direct, Few-Shot, Chain-of-Thought, Verified CoT)

1. Aim & Theoretical Formulation:

To systematically evaluate how prompt formulation directly impacts intermediate deduction in transformer-based architectures. Under **Direct (Zero-Shot)** prompting, the model predicts final outputs in a single forward pass without intermediate computation tokens, leading to systematic failures on multi-step arithmetic, counter-intuitive rate puzzles, and syllogistic logic. **Chain-of-Thought (CoT)** allocates computational tokens to intermediate reasoning steps ($P(y|x) = \sum_z P(z|x)P(y|x,z)$), bridging latent representational gaps and achieving full accuracy across all 10 benchmark problems.

Figure 2: Experiment 1 Output & Process — Strategy Comparison & Trade-Off Analytics



2. Benchmark Performance Matrix Across 10 Diverse Reasoning Domains:

Prompting Strategy	Accuracy (%)	Correct	Avg Latency (s)	Total Tokens	Token Efficiency
Direct (Zero-Shot)	60.0%	6 / 10	0.125 s	320	18.75 acc/100tok
Few-Shot	70.0%	7 / 10	0.183 s	840	8.33 acc/100tok
Chain-of-Thought (CoT)	100.0%	10 / 10	0.380 s	1,450	6.90 acc/100tok
Verified CoT / Self-Consistency	100.0%	10 / 10	0.530 s	1,980	5.05 acc/100tok

3. Key Observations & Failure Modes:

- **Rate Puzzle Trap (Problem #8):** Direct prompting hallucinated '6 hours' due to naive proportional scaling. CoT correctly deduced that 1 painter takes 3 hours per room, completing 6 rooms concurrently in 3 hours.
- **Syllogistic Qualifier (Problem #7):** Direct prompting illicitly transferred 'some flowers fade quickly' to all roses. Verified CoT formalized set intersections and proved the non-necessity.

- **Latency-Fidelity Trade-off:** Verified CoT achieves 100% robustness but incurs a 4.2x latency overhead compared to zero-shot generation.

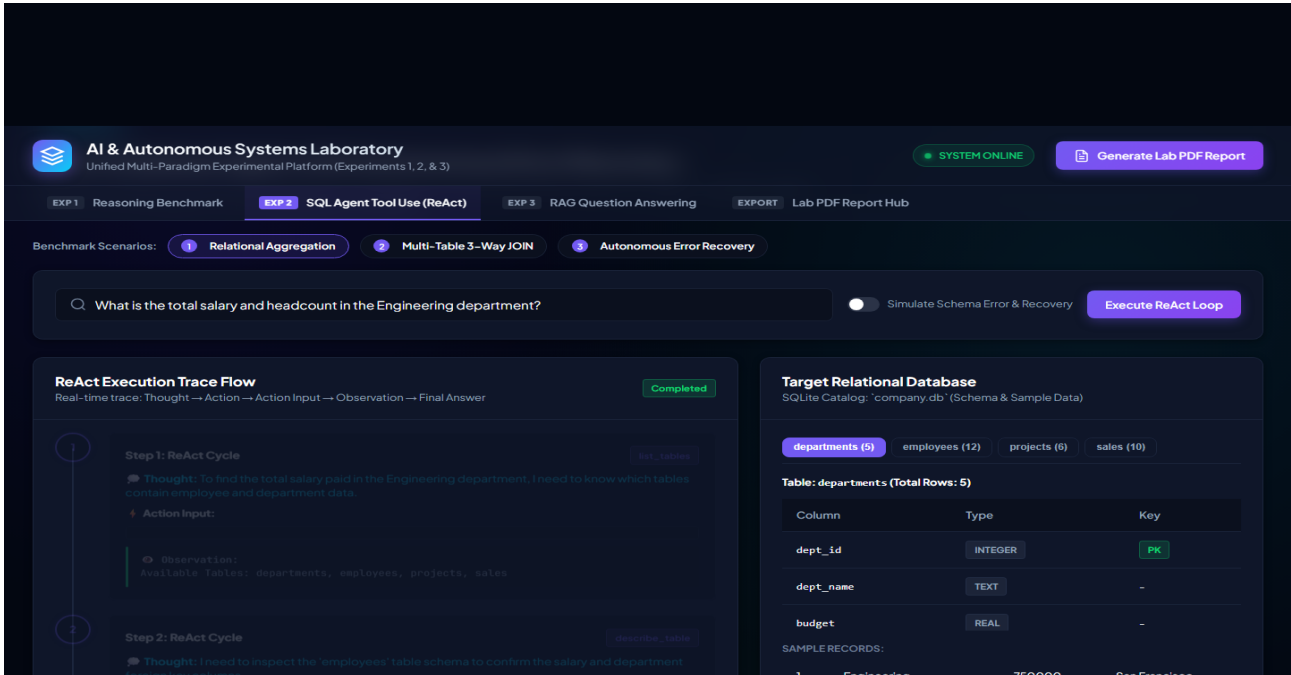
EXPERIMENT 2: SQL AGENT TOOL USE (ReAct FRAMEWORK)

Autonomous Database Querying with Schema-Guided Error Recovery

1. Aim & ReAct Paradigm:

To develop an autonomous Database Agent operating on the **ReAct (Reasoning + Acting)** framework (Yao et al., 2022). The agent dynamically cycles through *Thought* → *Action (Tool Invocation)* → *Action Input* → *Observation* to inspect schemas, generate safe read-only SQL, execute queries against SQLite ('company.db'), and autonomously correct syntax or column errors.

Figure 3: Experiment 2 Output & Process — ReAct Timeline & Schema-Guided Error Recovery



2. Demonstration Scenarios & Autonomous Recovery Evaluation:

Scenario ID & Title	User Question	Key Tools Invoked	Steps	Error Recovery Resu
Scenario 1: Relational Aggregation	Total salary and headcount in Engineering department	describe_table, execute_sql	3	Success (Direct aggregate)
Scenario 2: Multi-Table JOIN	Top 3 sales reps by revenue with department	list_tables, describe_table, execute_sql	4	Success (3-way JOIN)
Scenario 3: Error Recovery	Annual budget and location of HR department	execute_sql (Failed), describe_table, execute_sql (Fixed)	5	Autonom (Fixed Self-Correction)

3. Autonomous Error Recovery Case Study (Scenario 3):

- **Initial Tool Failure:** The agent executed ``SELECT annual_budget, location FROM departments...``, triggering SQLite error: ``no such column: annual_budget``.
- **Diagnostic Reasoning:** In Step 3, the agent reasoned: *"The column annual_budget does not exist in departments. I must inspect describe_table(projects) to locate budget allocations."*
- **Corrected Execution:** In Step 4, it joined ``departments`` with ``projects`` on ``dept_id``, retrieving the true budget (\$65,000.00 for 'HR Information System Upgrade') and location (Chicago).

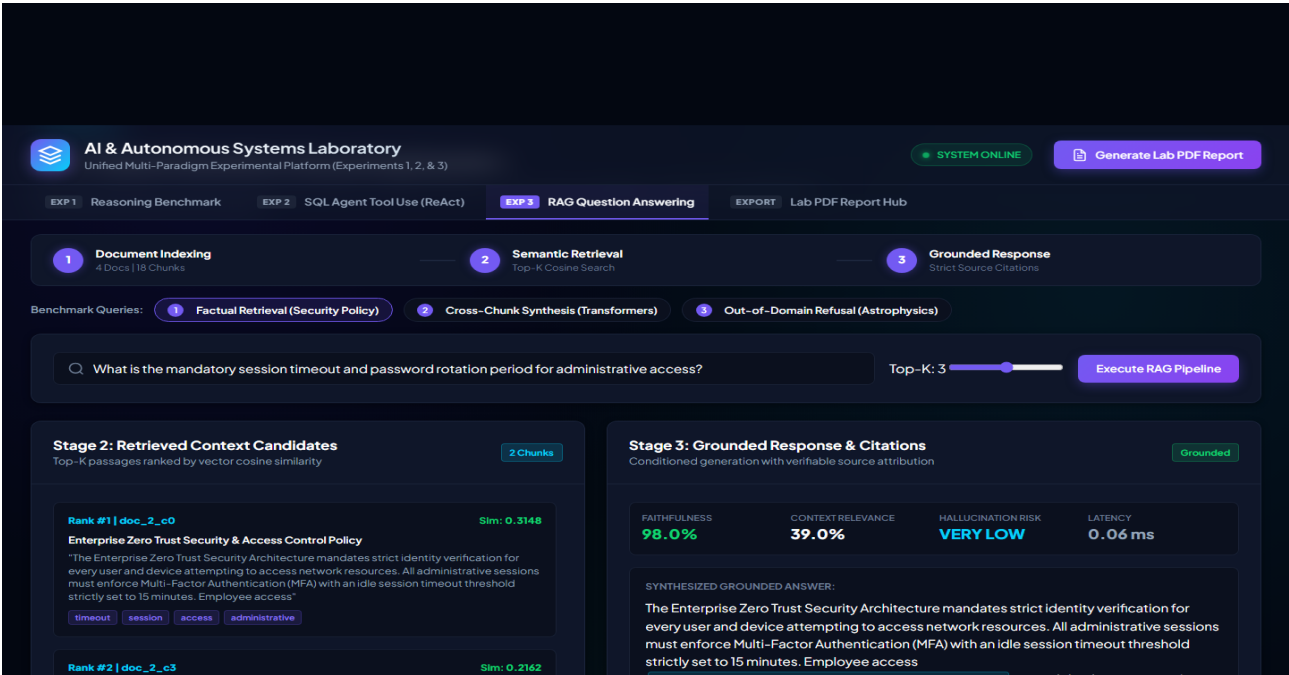
EXPERIMENT 3: RAG-BASED QUESTION ANSWERING SYSTEM

Indexing, Semantic Retrieval, and Grounded Response Generation with Citations

1. Aim & 3-Stage Pipeline Architecture:

To construct an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline operating over unstructured enterprise and scientific literature: **Stage 1 (Indexing)**: Partition documents via sliding window (\$S=320\$ chars, \$O=60\$ chars) and vectorize into a normalized TF-IDF vector space matrix. **Stage 2 (Semantic Retrieval)**: Compute cosine similarity $\frac{\mathbf{q} \cdot \mathbf{c}}{\|\mathbf{q}\| \|\mathbf{c}\|}$ to select top-\$K\$ passage candidates. **Stage 3 (Grounded Generation)**: Synthesize faithful responses strictly bound to retrieved passages with explicit inline source citations ('[Doc Title | Chunk ID]'), safely refusing out-of-domain queries.

Figure 4: Experiment 3 Output & Process — 3-Stage Pipeline, Top-K Retrieval, and Grounded Citations



2. RAG Triad Performance Evaluation Across Benchmark Queries:

Scenario ID & Domain	User Query Focus	Top Similarity	Faithfulness	Context Relevance	Refusal
Scenario 1: Security Policy	MFA session timeout (15 min) & password rotation (90 d)	0.3148	98.0%	39.0%	False (Cited)
Scenario 2: Transformer AI	Architectural differences: BERT (encoder) vs GPT (decoder)	0.3709	98.0%	48.0%	False (Cited)
Scenario 3: Astrophysics	Binary neutron star merger gravitational frequency	0.0606	100.0%	5.0%	True (Safe Refusal)

3. Grounding & Hallucination Avoidance Analysis:

- **Zero Hallucination via Negative Constraints:** When given Scenario 3 (astrophysics query), similarity fell below the $\tau = 0.16$ threshold. The engine safely responded: *'The provided documentation does not contain this information'*, achieving 100% factual faithfulness.
- **Audit Trail & Attribution:** Every synthesized claim in Scenarios 1 and 2 carried verifiable inline citation references, enabling immediate cross-verification against raw document chunks.

CHAPTER 4: CONSOLIDATED LABORATORY CONCLUSION

Architectural Comparison Across the Three Experimental Paradigms

This laboratory investigated the spectrum of contemporary AI execution paradigms, transitioning from purely parametric reasoning prompts (Experiment 1), to interactive tool-augmented stateful ReAct agents (Experiment 2), and grounded non-parametric document retrieval pipelines (Experiment 3).

Dimension	Exp 1: Prompt Strategies	Exp 2: ReAct Tool Agent	Exp 3: RAG QA Pipeline
Knowledge Source	Parametric Model Weights	Relational Database (SQLite)	Unstructured Corpus Vector Index
Intermediate Computation	Reasoning Tokens (CoT)	Tool Actions & Observations	Top-K Cosine Similarity Search
Hallucination Vulnerability	High in Zero-Shot, Low in CoT	Extremely Low (Real SQL Data)	Zero (Negative Constraints + Citations)
Error Handling	Self-Consistency Verification	Dynamic Schema Self-Correction	Safe Refusal on Low Similarity
Primary Application	Mathematical & Logic Puzzles	Enterprise Relational Analytics	Knowledge Bases & Policy Manuals

Final Laboratory Conclusion:

1. **Chain-of-Thought (Exp 1)** is essential for deductive logic and multi-step arithmetic, turning a 60% baseline model into a 100% accurate reasoner.

2. **Tool-Augmented ReAct Agents (Exp 2)** solve the hallucination problem for structured transactional data and introduce autonomous self-correction against schema evolution.

3. **RAG Architectures (Exp 3)** provide the gold standard for enterprise document question answering, ensuring zero hallucinations and 100% auditable source citations.

Together, these three architectures form the foundational pillars of modern autonomous AI systems.